

Module-13 : Networking with Windows Server

Q.1 Discuss the role of Windows Firewall in Windows Server and how to configure it.

ANS:

Windows Firewall is a built-in security feature that protects your server by filtering network traffic entering and exiting based on defined security rules. It acts as a barrier between your server and external traffic, preventing unauthorized access and attacks.

Key Functions

- Filters incoming and outgoing packets based on security rules
- Blocks unauthorized access attempts
- Allows administrators to create custom rules for specific applications/ports
- Supports three profile types: Domain, Private, and Public

Configuration Steps

1. **Open Windows Firewall:** Server Manager → Tools → Windows Firewall with Advanced Security
2. **Create Inbound/Outbound Rules:** Right-click "Inbound Rules" or "Outbound Rules" → New Rule
3. **Specify Rule Type:** Choose Program, Port, Predefined, or Custom
4. **Set Action:** Allow, Allow if secure, or Block
5. **Configure Profiles:** Select when rule applies (Domain/Private/Public)
6. **Name and Complete:** Provide descriptive name and finish wizard

Q.2 What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

ANS: **What is NAT?**

NAT is an IP translation and mapping protocol that works at the network layer (Level 3) of the OSI model. It allows packets from a private network to be routed to the Internet by translating private IP addresses to a public IP address.

Configuration Steps (Windows Server 2019/2016)

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1. **Install RRAS Role:** Server Manager → Add Roles and Features → Routing and Remote Access Service
2. **Open RRAS Console:** Tools → Remote and Routing Access
3. **Configure RRAS:** Right-click server name → Configure and Enable Routing and Remote Access
4. **Select Configuration:** Choose "Network Address Translation (NAT)" → Next
5. **Select Internet Interface:** Choose the network interface connecting to the internet
6. **Finish Setup:** Click Finish to complete wizard
7. **Verify Interfaces:**
 - LAN interface = Private interface
 - Internet interface = Public interface with "Enable NAT" checked
8. **Test:** Ping 8.8.8.8 and access www.google.com

Q.3 Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

ANS: **Concept**

DHCP automatically assigns IP addresses and network configuration parameters to devices on a network, eliminating manual IP configuration.

Key Components

- **Scope:** Range of IP addresses available for assignment
- **Lease:** Duration a client can use an IP address
- **Reservation:** Permanent IP assignment for specific devices
- **Options:** DNS servers, gateway, WINS servers

Configuration in Windows Server 2016

1. **Install DHCP Role:** Server Manager → Add Roles and Features → DHCP Server
2. **Open DHCP Manager:** Tools → DHCP
3. **Create New Scope:** Right-click IPv4 → New Scope

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4. Scope Wizard:

- Name and description (e.g., "Administration Building Clients")
- IP address range (e.g., 192.168.1.100-200)
- Subnet mask (e.g., 255.255.255.0)
- Exclusions (addresses to skip)
- Lease duration (default: 8 days)

5. Configure Options:

- Router (Default Gateway)
- DNS servers
- Domain name

6. **Activate Scope:** Right-click new scope → Activate

7. **Verify:** Use PowerShell to confirm scope is active

Q.4 Describe the process of configuring DNS (Domain Name System) in Windows Server.

ANS:

Configuration Steps

1. **Install DNS Role:**

- Open Server Manager
- Click Add roles and features
- Select DNS Server from roles list → Complete installation

2. **Create Zone:**

- Open DNS Manager (Tools → DNS)
- Right-click server name → New Zone
- Choose Primary Zone → Next
- Specify zone name (e.g., yourdomain.com)

3. **Create Resource Records:**

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- Right-click new zone → New Host (A or AAAA)
 - Enter name (e.g., www) and IP address
 - Click Add Host
 - Repeat for MX, CNAME, PTR records
4. **Configure Forwarders:**
- Right-click server → Properties → Forwarders tab
 - Click Edit → Enter ISP DNS or public DNS (e.g., 8.8.8.8)
5. **Configure Root Hints:** Ensure properly configured for fallback if forwarders fail.

Q.5 What is Server Manager, and how do you use it to manage servers in Windows Server?

ANS: **What is Server Manager?**

Server Manager is a single management console that provides a unified interface for managing local and remote Windows Servers. It's enabled by default when Windows Server is installed.

Key Features

- Single source for managing identity and system information
- Dashboard with server health and performance overview
- Add/remove roles and features
- Manage multiple servers simultaneously
- Event data configuration and service management
- Best Practices Analyzer (BPA)
- Create server groups

How to Use It

Windows Server Manager

1. **Open Server Manager:** Start menu → Server Manager
2. **Add Servers:** Manage → Add Servers → Find Now → Select servers → OK
3. **Add Roles/Features:** Manage → Add Roles and Features

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4. **View Dashboard:** Shows local server configuration, roles, storage, and performance
5. **Manage Multiple Servers:** All Servers view shows all managed servers
6. **Monitor Performance:** Performance panel shows CPU, memory, disk usage

Q.6 Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it.

ANS: Role in Windows Server 2016/2019

RDS enables users to access virtual desktops and applications remotely, providing centralized application delivery and remote workspace capabilities.

Key Components

- **RD Connection Broker:** Manages user sessions and load balancing
- **RD Session Host:** Hosts remote applications and desktops
- **RD Web Access:** Provides web portal for accessing resources
- **RD Gateway:** Enables secure external access over HTTPS
- **RD Licensing:** Manages RDS CALs (Client Access Licenses)

Configuration Steps (Standard Deployment)

1. **Add Servers to Server Manager:**
 - Manage → Add Servers → Find Now → Select deployment servers → OK
2. **Deploy RDS:**
 - Manage → Add Roles and Features
 - Select Remote Desktop Services installation → Standard Deployment → Session-based desktop deployment
 - Select servers for RD Connection Broker, RD Web Access, RD Session Host
 - Check "Restart automatically if required" → Deploy
3. **Add RD License Server:**
 - Remote Desktop Services → Overview → +RD Licensing
 - Select server → Add

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4. Activate License Server:

- Remote Desktop Services → Servers → Right-click server → RD Licensing Manager
- Action → Activate Server → Follow wizard

5. Add RD Gateway:

- Remote Desktop Services → Overview → +RD Gateway
- Select server → Enter SSL certificate name (external FQDN) → Add

6. Create Certificates:

- Overview → Tasks → Edit Deployment Properties
- Certificates → RD Gateway → Create new certificate
- Enter FQDN and password → Save

7. Configure Deployment Properties:

- Clear "Bypass RD Gateway for local addresses"
- Select "Per User" licensing

8. Create Session Collection:

- Collections → Tasks → Create Session Collection
- Enter name → Select RD Session Host → Set profile disk path → Create